AGENTS FOR IMPACT 2026 · CHALLENGE 2

Atlanta Food Rescue Broker

An agent that moves a grocer’s perishable surplus to a pantry that can actually keep it cold, before the clock runs out.

Track: Retail & Grocer Rescue · **Metro:** Atlanta

[Team name] · [Presenter names]

OUR SCENARIO

“We coordinate food rescue in Atlanta. A Midtown grocer has just offered us 10 crates of spinach that must stay cold and leave the dock by 8:10 PM. Here is who we’d have to call, and here is what we don’t know about them.”

Retail rescue means small, mixed, perishable offers going to pantries that often have only a household fridge. That makes it a different problem from moving pallets between warehouses.

WHAT WE BUILT

From a grocer’s post to a drafted call

Grocer posts
In their own words, from a phone

→

Gemini agent
Google ADK, reads the offer and the clock

→

Vector search
BigQuery, 3,072-dim Gemini embeddings

→

Filter and rank
Storage, capacity, open days, the clock, need

→

Match Brief
Who, why, the clock, the message

Google-managed MCP
BigQuery’s managed MCP server and MCP Toolbox for Databases

Offer state on Cloud SQL
Remembers every offer, reply and hold

Deployed on Cloud Run
Agent, offer state, console and phone screens

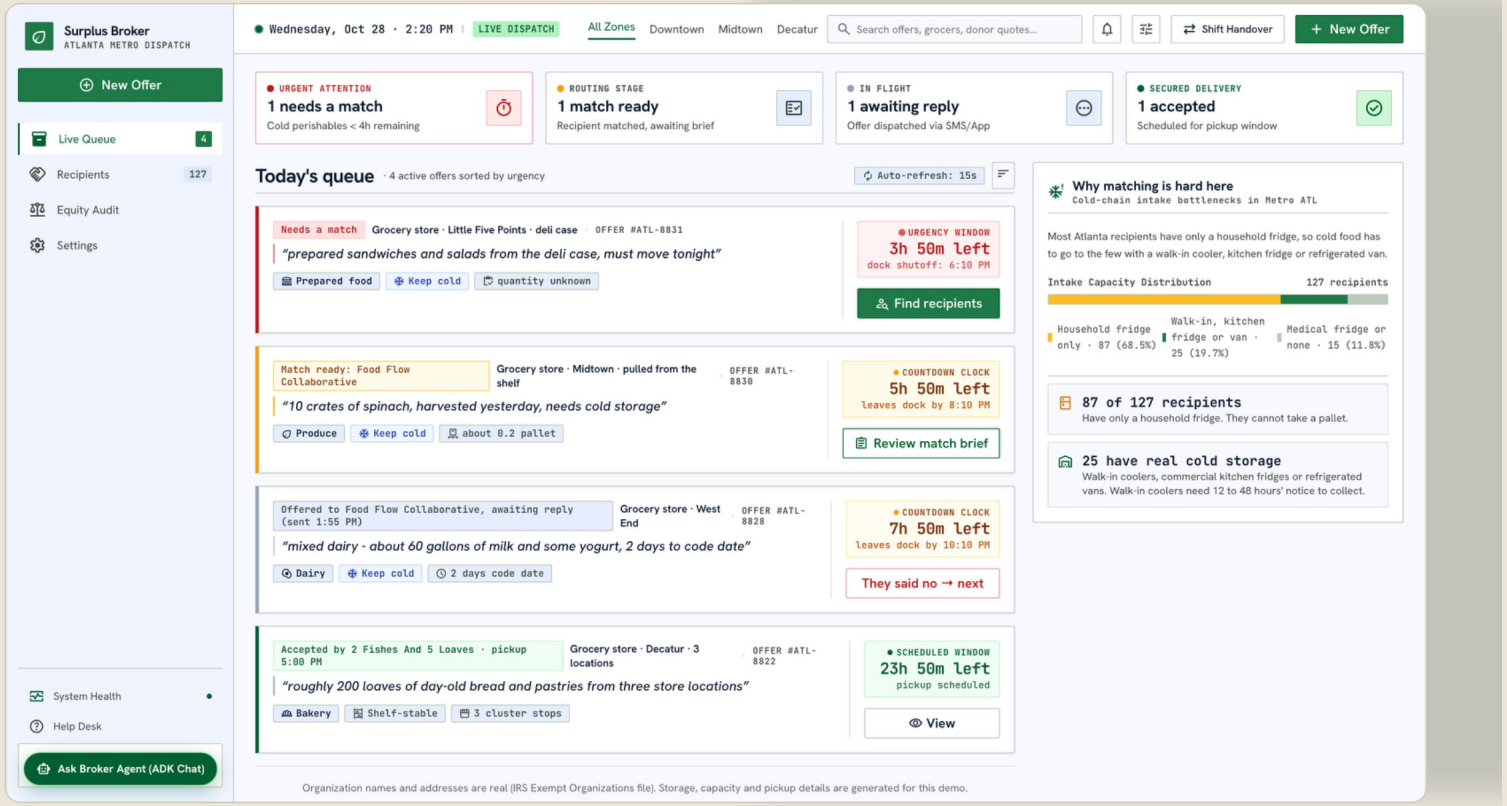
THE COORDINATOR CONSOLE

Every offer, sorted by its clock

A countdown on every offer to the moment it has to leave the dock.

Status at a glance: needs a match, match ready, awaiting reply, accepted.

Why it's hard here: 87 of 127 Atlanta recipients have only a household fridge.



THE MOMENT THAT MATTERS

Greenline Community Pantry is the 2nd closest match by meaning. It has no refrigerator.

Its profile asks for fresh greens, so vector search loves it. The spinach would spoil there. Our agent rules it out and says why.

THE OUTPUT: A MATCH BRIEF

Ready to act on, no follow-up questions

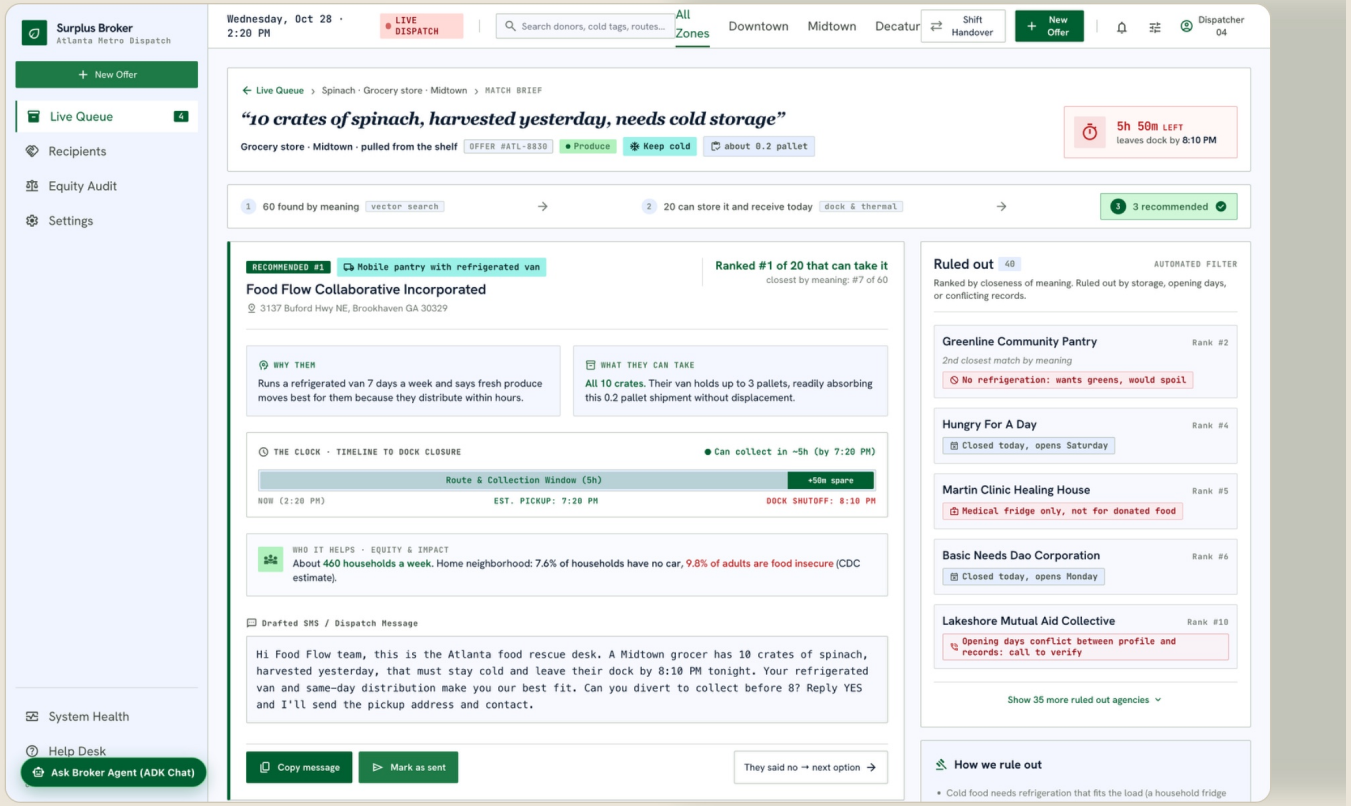
60 → 20 → 3

found by meaning, can store it today, recommended

Who, why, what they can take, the clock with 50 min spare, who it helps, and a drafted text

Ruled out, with reasons: Greenline at rank #2, no fridge

If they say no: next option, or split 5 + 5 crates to higher-need pantries



THE “THEY SAID NO” PROBLEM

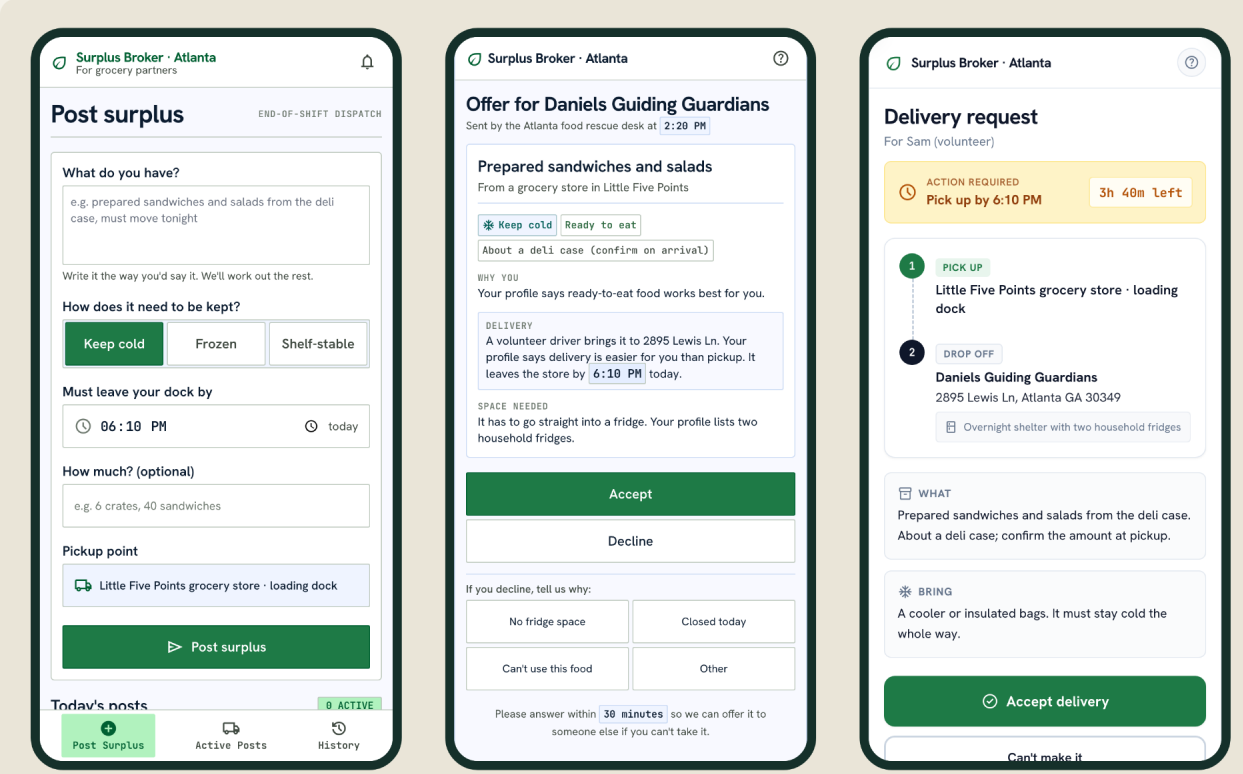
One offer, seen by everyone it touches

Grocer posts in their own words, with a must-leave-by time.

Pantry accepts, or declines with a one-tap reason.

Driver accepts the pickup when the pantry has no vehicle.

Cloud SQL remembers it all: a decline sticks, and a pantry with a live offer can’t be double-booked.



EQUITY AUDIT · ALL TEN POSTINGS

Did the food go where the need is?

Where the food goes	No vehicle	Poverty	Need pctile	To planted rows	Pallets placed
Atlanta metro, population-weighted	12.9%	18.5%	0.47	–	–
Pool: every generated recipient	13.4%	18.0%	0.48	–	–
SQL matcher, fixed need weight, ZIP tracts	19.8%	24.5%	0.65	27%	41.7
SQL matcher, clock-scaled weight, geocoded	26.4%	31.8%	0.77	20%	23.7
Agent matcher (agent/tools.py)	19.0%	29.4%	0.72	50%	20.7

Yes. Our final matcher (highlighted) sends food to tracts with about **twice the metro's no-vehicle rate**. Need means poverty, vehicle access and assistance only; demographic composition is never an input.

WHAT THE AUDIT TAUGHT US

Four findings, including two we don’t like

The pool looks like the metro

Recipients aren’t clustered in high-need tracts, so any skew comes from the matcher.

Geocoding changed the answer

93 of 109 matched addresses landed in a different tract than their ZIP centroid. Need percentile rose from 0.65 to 0.77.

Tradeoff: need versus volume

On the 30-pallet squash, need weighting plus a 3-stop cap places 3 pallets, not 21. Small high-need pantries can’t absorb farm-sized loads.

Failure mode: invented locations

The agent matcher sent half its food to planted teaching rows. Their made-up coordinates sit in a high-need tract, and need weighting rewarded that.

WHICH HALF OF OUR DATA IS REAL

Real organizations, generated operations

Real

- **Organizations:** IRS Exempt Organizations file. Real names, addresses and reported income
- **Shelf life:** USDA FoodKeeper, 661 products
- **Need:** American Community Survey tracts; 109 of 118 addresses geocoded by the Census Bureau

Generated

- **Operational profiles:** cold storage, pallets, pickup windows. No pantry publishes these
- 14 archetypes, seeded by tax ID, phrased like real Seattle and Pennsylvania profiles
- **9 planted teaching rows**, labeled in every brief

Limits: FoodKeeper is home-storage guidance, not a safety guarantee. Collection times are estimates, not live drive times. Pallet sizes are estimated. 9 addresses fall back to the ZIP centroid.

WHAT WE'D DO NEXT

Make the generated half real

Pantries confirm their profiles

Each "yes" or "no" teaches us what storage they really have.

Real drive times

Maps routing in place of collection-window estimates.

Need without the volume penalty

Split large loads across more high-need stops.

The question to ask of any match: did the food go where the need is, or where the loading docks are?

github.com/xolotlatoani/A4I2026-challenge-2-food-equity · [Cloud Run URL]